

FITBUDDY AI

Phase 3: Project Design Phase

Prepared from the current public FitBuddy_ai repository and organized according to the Phase 1–6 structure of the provided AI/ML & GenAI project template.

3.1 System Architecture

FitBuddy follows a lightweight web application architecture. The browser submits form data to FastAPI. The route layer constructs the plan, stores it through the SQLite database layer, and renders Jinja2 templates for the result. Feedback follows the same route/database/template cycle.

3.2 Architecture Flow

Browser UI → FastAPI application → Route handlers → SQLite database → Jinja2 dashboard
■ Feedback route → updated_plan

3.3 Main Components

Component	Responsibility
main.py	Creates the FastAPI application, initializes the database at startup, and includes the router.
routes.py	Implements home, plan generation, user dashboard, feedback and admin routes.
database.py	Defines SQLite location, creates the users table and returns database connections.
config.py	Defines application settings, environment-file loading and Gemini model configuration.
index.html	Provides the profile registration interface.
dashboard.html	Displays profile, nutrition tip, workout plan and feedback form.
admin.html	Displays all stored user and plan records.

3.4 Route Design

Route	Purpose
GET /	Render the FitBuddy registration page.
POST /generate-plan	Receive profile data, create a plan, save it, and render the dashboard.
GET /user/{user_id}	Retrieve one profile and render its dashboard.
POST /submit-feedback	Store feedback and create the updated plan record.
GET /users	Retrieve all user records and render the admin registry.

3.5 Database Design

Field	Type	Purpose
id	INTEGER	Primary key.

Field	Type	Purpose
name	TEXT	User name.
age	INTEGER	Age value.
gender	TEXT	Gender selection.
weight	REAL	Weight in kg.
height	REAL	Height in cm.
days	INTEGER	Workout days per week.
goal	TEXT	Fitness objective.
level	TEXT	Experience level.
original_plan	TEXT	Initial generated blueprint.
updated_plan	TEXT	Feedback-adjusted plan.
feedback_text	TEXT	User feedback.

3.6 GenAI Design Extension

The configuration file contains Gemini API/model settings. A future production design can route the validated profile into a Gemini workout-generation service and a separate nutrition/recovery service. The current public routes, however, construct the workout blueprint directly, so the architecture should treat Gemini integration as an extension rather than claim that the current route already performs that call.

3.7 UI Design

- Registration page: centered FitBuddy AI identity, profile fields, objective and experience selectors.
- Dashboard: profile metrics, nutrition tip, active workout schedule, baseline plan and feedback form.
- Admin page: tabular display of users, metrics, goals, original plans, updated plans and feedback.

Project repository: https://github.com/asvikamoorthy-cyber/FitBuddy_ai

Reference structure: <https://github.com/Ravi-teja-777/AI-ML-and-GEN-AI-Track-Project-Template>

Scope note: This document describes the project state visible in the public repository and identifies proposed/future items separately where applicable.